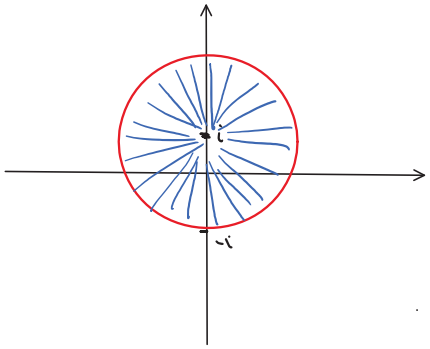
Problema 1 (tercer parcial)

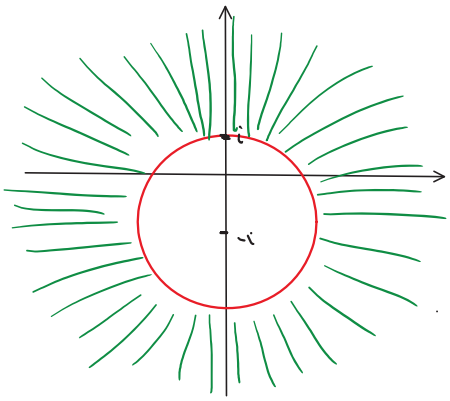
domingo, 22 de diciembre de 2024 19:59

Obtén al menos los primeros cuatro elementos del desarrollo de Laurent de $f(z) = \dfrac{1}{(z-i)^2(z+i)}$ en los siguientes conjuntos:

a) [1.25 puntos] $0 < |z-i| < 2$

b) [1.25 puntos] $|z+i| > 2$





a) $|z-i| > 0$

$|z-i| < 2 \Rightarrow \left|\frac{z-i}{2}\right| < 1$

$$f(z) = \frac{1}{(z-i)^2(z+i)} = \frac{1}{(z-i)^2} \cdot \frac{1}{z-i+2i} = \frac{1}{(z-i)^2} \cdot \frac{1}{2i\left(\frac{z-i}{2i}+1\right)} =$$

$$= \frac{1}{(z-i)^2} \cdot \frac{1}{2i} \cdot \frac{1}{1+\left(\frac{z-i}{2i}\right)} = \frac{1}{2i} \frac{1}{(z-i)^2} \cdot \frac{1}{1-\left(-\left(\frac{z-i}{2i}\right)\right)} =$$

$\uparrow$ $\left|\frac{z-i}{2i}\right| = \left|\frac{z-i}{2}\right| < 1$

$$= \frac{1}{2i} \frac{1}{(z-i)^2} \cdot \sum_{n=0}^{\infty} \left(-\left(\frac{z-i}{2i}\right)\right)^n = \frac{1}{2i} \frac{1}{(z-i)^2} \cdot \left(1 - \left(\frac{z-i}{2i}\right) + \left(\frac{z-i}{2i}\right)^2 - \left(\frac{z-i}{2i}\right)^3 + \cdots\right) =$$

$$= \frac{1}{2i} \cdot \frac{1}{(z-i)^2} - \frac{1}{(2i)^2} \frac{1}{z-i} + \frac{1}{(2i)^3} - \frac{1}{(2i)^4}(z-i) + \frac{1}{(2i)^5}(z-i)^2 - \cdots$$

$$= \boxed{\frac{-i}{2}} \frac{1}{(z-i)^2} + \boxed{\frac{1}{4}} \frac{1}{z-i} + \boxed{\frac{i}{8}} \boxed{-\frac{1}{16}}(z-i) \boxed{-\frac{i}{32}}(z-i)^2 - \cdots$$

b) $|z+i| > 2 \Rightarrow \left|\frac{2}{z+i}\right| < 1$

$$f(z) = \frac{1}{(z-i)^2(z+i)} = \frac{1}{(z+i)} \cdot \left(\frac{1}{z-i}\right)^2 = \frac{1}{z+i}\left(\frac{1}{z+i-2i}\right)^2 =$$

$$= \frac{1}{z+i}\left(\frac{1}{(z+i)\left(1-\frac{2i}{z+i}\right)}\right)^2 = \frac{1}{(z+i)}\frac{1}{(z+i)^2}\left(\frac{1}{1-\left(\frac{2i}{z+i}\right)}\right)^2 =$$

$$\uparrow \quad \left|\frac{2i}{z+i}\right| = \left|\frac{2}{z+i}\right| < 1$$

$$= \frac{1}{(z+i)^3}\left(\sum_{n=0}^{\infty}\left(\frac{2i}{z+i}\right)^n\right)^2 = \frac{1}{(z+i)^3}\left(1 + \frac{2i}{z+i} + \frac{(2i)^2}{(z+i)^2} + \frac{(2i)^3}{(z+i)^3} + \cdots\right)^2 =$$

$$= \frac{1}{(z+i)^3}\left(1 + \frac{2i}{z+i} + \frac{(2i)^2}{(z+i)^2} + \frac{(2i)^3}{(z+i)^3} + \cdots\right)\left(1 + \frac{2i}{z+i} + \frac{(2i)^2}{(z+i)^2} + \frac{(2i)^3}{(z+i)^3} + \cdots\right) =$$

$$= \frac{1}{(z+i)^3}\left(1 + (2i+2i)\cdot\frac{1}{z+i} + \left((2i)^2+(2i)^2+(2i)^2\right)\frac{1}{(z+i)^2} + \left((2i)^3+(2i)^3+(2i)^3+(2i)^3\right)\frac{1}{(z+i)^3} + \cdots\right)$$

$$= \frac{1}{(z+i)^3} + 4i\frac{1}{(z+i)^4} + 3\cdot(2i)^2\cdot\frac{1}{(z+i)^5} + 4\cdot(2i)^3\frac{1}{(z+i)^6} + \cdots =$$

$$= \boxed{1}\cdot\frac{1}{(z+i)^3} + \boxed{4i}\frac{1}{(z+i)^4} \boxed{-12}\frac{1}{(z+i)^5} \boxed{-32i}\frac{1}{(z+i)^6} + \cdots$$

Problema 2 (tercer parcial)

domingo, 22 de diciembre de 2024 19:59

Determina la función holomorfa $f(z)$ que cumple las condiciones $\mathrm{Re}[f'(z)] = 3(x^2 - y^2) - 4y$, $f(1+i) = 0$, $f'(0) = 0$.

$g(z) = f'(z) = u(x,y) + i\cdot v(x,y) = \left(3(x^2-y^2) - 4y\right) + i\cdot v(x,y)$

① Vamos a comprobar que $u(x,y)$ es una función armónica:

$u_x(x,y) = 6x \longrightarrow u_{xx}(x,y) = 6$

$v_y(x,y) = -6y - 4 \longrightarrow v_{yy}(x,y) = -6$

$u_{xx}(x,y) + v_{yy}(x,y) = 6 - 6 = 0$

② A continuación calcularemos la función $g(z)$ holomorfa.

$u_x(x,y) = v_y(x,y) \Rightarrow v_y(x,y) = 6x \Rightarrow v(x,y) = \int 6x\,dy + C(x) = 6xy + C(x)$

$u_y(x,y) = -v_x(x,y) \Rightarrow -6y - 4 = -\left(6y + C'(x)\right) \Rightarrow C'(x) = 4 \Rightarrow C(x) = 4x + C$

$\Rightarrow v(x,y) = 6xy + 4x + C \Rightarrow g(z) = \left(3(x^2-y^2) - 4y\right) + i\left(6xy + 4x + C\right)$

$\Rightarrow g(z) = 3z^2 + i\cdot(4z + C)$

$g(0) = f'(0) = 0 \Rightarrow 0 + i\cdot C = 0 \Rightarrow C = 0 \Rightarrow g(z) = 3z^2 + i\,4z$

③ $g(z) = f'(z) \Rightarrow f(z) = \int g(z)\,dz = \int (3z^2 + i4z)\,dz = z^3 + i\cdot 2z^2 + C$

$f(1+i) = 0 \Rightarrow (1+i)^3 + i\cdot 2\cdot(1+i)^2 + C = 0 \Rightarrow$

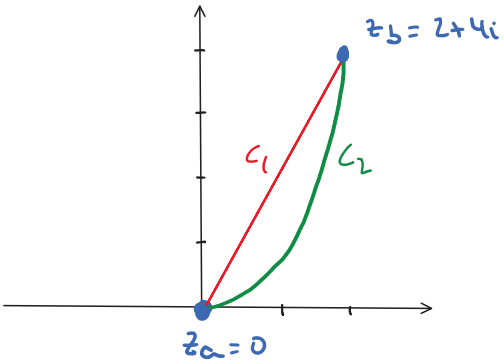
$\Rightarrow (1 + 3i + 3i^2 + i^3) + i\cdot 2(1 + 2i + i^2) + C = 0 \Rightarrow (-2 + 2i) + i\cdot 2\cdot(2i) + C = 0 \Rightarrow$

$\Rightarrow C = 2 - 2i + 4 = 6 - 2i \Rightarrow \boxed{f(z) = z^3 + i\cdot 2z^2 + (6 - 2i)}$

Problema 3 (tercer parcial)

domingo, 22 de diciembre de 2024 19:59

Calcula la integral $\int_C (8x^2 - iy^3)dz$ entre los puntos $z_a = 0$ y $z_b = 2+4i$ a través de dos trayectorias distintas y comenta los resultados.



C_1: $\alpha(t) = (t, 2t) \to z(t) = \overset{x(t)}{t} + i\overset{y(t)}{2t}$

$t_a = 0,\ t_b = 2$ $\qquad z'(t) = 1 + 2i$

C_2: $\alpha(t) = (t, t^2) \to z(t) = \overset{x(t)}{t} + i\overset{y(t)}{t^2}$

$t_a = 0,\ t_b = 2$ $\qquad z'(t) = 1 + i2t$

- Curva C_1:

$$\int_C (8x^2 - iy^3)\,dz = \int_0^2 \left(8\cdot t^2 - i(2t)^3\right)\cdot(1+2i)\,dt = (1+2i)\int_0^2 (8t^2 - i\cdot 8t^3)\,dt =$$

$$= (1+2i)\cdot 8\left[\frac{t^3}{3} - i\frac{t^4}{4}\right]_0^2 = 8(1+2i)\cdot\left(\frac{8}{3} - i\cdot\frac{16}{4}\right) = \boxed{\frac{256}{3} + i\,\frac{32}{3}}$$

- Curva C_2:

$$\int_C (8x^2 - iy^3)\,dz = \int_0^2 \left(8t^2 - i(t^2)^3\right)\cdot(1+i2t)\,dt = \int_0^2 (8t^2 - it^6)\cdot(1+i2t)\,dt =$$

$$= \int_0^2 \left(8t^2 + i\cdot 16t^3 - it^6 + 2t^7\right)dt = \int_0^2 \left(\left(8t^2 + 2t^7\right) + i\left(16t^3 - t^6\right)\right)dt =$$

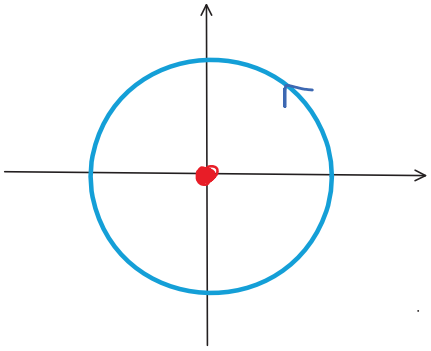
$$= \left[\left(\frac{8t^3}{3} + \frac{t^8}{4}\right) + i\left(4t^4 - \frac{t^7}{7}\right)\right]_0^2 = \left(\frac{64}{3} + \frac{256}{4}\right) + i\left(64 - \frac{128}{7}\right) = \boxed{\frac{256}{3} + i\,\frac{320}{7}}$$

Al no ser $f(z) = 8x^2 - iy^3$ una función holomorfa (no cumple las ecuaciones Cauchy-Riemann), es de esperar que la integral dependa del camino.

Problema 4 (tercer parcial)

domingo, 22 de diciembre de 2024 19:59

Calcula la integral de $f(z) = \left(z^2 + \frac{1}{z}\right) \cos\left(\frac{1}{z}\right)$ a lo largo de la curva $|z| = 1$ recorrida en sentido positivo.



Puesto que la curva contiene en su interior una singularidad, vamos a utilizar el Teorema de los residuos.

$$\oint_C f(z)\,dz = 2\pi i \cdot \text{Res}(f, z=0)$$

$$\cos(z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n} = 1 - \frac{1}{2}z^2 + \frac{1}{24}z^4 - \frac{1}{720}z^6 + \dots$$

$$\cos\left(\frac{1}{z}\right) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \left(\frac{1}{z}\right)^{2n} = 1 - \frac{1}{2}\frac{1}{z^2} + \frac{1}{24}\frac{1}{z^4} - \frac{1}{720}\frac{1}{z^6} + \dots$$

$$f(z) = \left(z^2 + \frac{1}{z}\right) \cdot \cos\left(\frac{1}{z}\right) = \left(z^2 + \frac{1}{z}\right) \cdot \left(1 - \frac{1}{2}\frac{1}{z^2} + \frac{1}{24}\frac{1}{z^4} - \frac{1}{720}\frac{1}{z^6} + \dots\right) =$$

$$= \left(z^2 - \frac{1}{2} + \frac{1}{24}\frac{1}{z^2} - \frac{1}{720}\frac{1}{z^4} + \dots\right) + \left(\frac{1}{z} - \frac{1}{2}\frac{1}{z^3} + \frac{1}{24}\frac{1}{z^5} - \frac{1}{720}\frac{1}{z^7} + \dots\right) =$$

$$= \left(z^2 - \frac{1}{2} + \boxed{\frac{1}{z}} + \frac{1}{24}\frac{1}{z^2} - \frac{1}{2}\frac{1}{z^3} - \frac{1}{720}\frac{1}{z^4} + \frac{1}{24}\frac{1}{z^5} - \dots\right)$$

$\hookrightarrow \text{Res}(f, z=0) = C_{-1} = 1$

Por lo tanto, $\oint_C f(z)\,dz = 2\pi i \cdot \text{Res}(f, z=0) = 2\pi i \cdot 1 = \boxed{2\pi i}$